VISHWAJIT MALL

Full Stack Developer | React, Node.js, Python, AI Applications

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PROFESSIONAL SUMMARY

Full Stack Developer with **5+ years** of experience shipping web, desktop, and mobile products with React, Next.js, Vue.js, React Native, Electron, Node.js (NestJS, Express.js), Python (FastAPI), and Laravel on PostgreSQL, MySQL, MongoDB, and Redis. Sole backend developer of FLO at SPNX (NestJS, 244 REST operations) and author of the OpenAPI contract its React frontend is built on. Builds AI products: RAG chat with citations, AI code review, natural-language automation, and photo-based nutrition tracking, with LLM fallback across OpenAI, Gemini, Groq, and Ollama.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, PHP, SQL, HTML5, CSS3
Frontend: React 19, Next.js 16, Vue.js, Angular, Vite, Tailwind CSS, Bootstrap, Zustand, React Query, responsive UI
Mobile & Desktop: React Native (Expo), Electron, Progressive Web Apps, offline-first sync
Backend: Node.js, NestJS, Express.js, FastAPI, Laravel, REST API design, GraphQL, WebSockets, WebRTC, OpenAPI/Swagger
AI / LLM: LLM integration, RAG, LangChain, LangGraph, AI agents with tool calling, Model Context Protocol (MCP), vision LLMs, vector search (Qdrant, pgvector), prompt engineering, LLM fallback (OpenAI, Gemini, Groq, Ollama)
Databases & ORMs: PostgreSQL, MySQL, MongoDB, Redis, Qdrant, pgvector, Prisma, SQLAlchemy 2, Eloquent
Auth & Security: JWT, OAuth 2.0 / OIDC (Microsoft Entra ID, Keycloak), RBAC, CSRF, rate limiting, AES-256-GCM
Cloud & DevOps: Docker, GitHub Actions, GitLab CI, Nginx, AWS (S3, RDS, Amplify), Azure (Blob, Key Vault), Shopify API
Testing & Tools: Jest, pytest, node:test, Playwright, Git, GitHub, JIRA, Azure DevOps, Agile/Scrum, Claude Code

PROFESSIONAL EXPERIENCE

Backend Developer | SPNX, India **Oct 2026 – Present**

FLO – VAT Registration Platform | *Node.js, NestJS 11, TypeScript, PostgreSQL 17, Prisma 7, Microsoft Entra ID, Azure, Jest*

- Sole backend developer (100 of 101 commits) of FLO, a staff workspace and client portal that runs EU VAT registrations for overseas businesses: **244 REST operations**, 34 NestJS modules, 63 Prisma models.
- Designed a data-driven workflow engine in pure TypeScript that runs each country's registration process as versioned data; the same guards enforce business rules and return each user's allowed actions.
- Implemented Microsoft Entra ID (OIDC) sign-in, Postgres-backed sessions with CSRF protection, and RBAC with 4 roles and 21 permissions; separate portal serializers keep internal data out of client responses.
- Integrated Azure Blob Storage, Azure Key Vault (AES-256-GCM for stored credentials), ClamAV / Defender virus scanning, Microsoft Graph e-mail, and Adobe Acrobat Sign behind ports and adapters.
- Published a code-first OpenAPI/Swagger contract from which the React 19 frontend team generates typed API clients.

Full Stack Developer | Bee Logical Software Solutions, Pune, India **Nov 2022 – Jun 2026**

- Core backend developer on 5 production platforms: designed APIs, data models, and auth, and worked with AI/ML engineers to ship LLM features. Received the **Outstanding Performance Award (Nov 2024)**.

Bella Canvas – 3D Product Customization E-Commerce | *Laravel, Vue.js, MySQL (AWS RDS), AWS S3, Shopify API*

- Led backend design for a 5-stage product design lifecycle: REST APIs for designs and assets, Shopify order sync, and an S3 pipeline serving textures for real-time 3D rendering.

SetBot – Property Booking Platform | *Node.js, Express.js, TypeScript, MySQL, Angular*

- Built REST APIs for listings, bookings, and admin reports for studios, hostels, restaurants, and shops, with JWT auth and RBAC across 4 roles.
- Built a booking engine with real-time slot validation and offer rules that stay correct under concurrent requests.

SmartHR – AI Interview Platform | *Python, FastAPI, LangChain, LangGraph, RAG*

- Built an AI interview pipeline with LangChain for LLM orchestration and LangGraph for stateful, multi-turn interviews.
- Built RAG over job descriptions to generate role-specific questions, and agent-based scoring that writes candidate reports.

Havu – Events-Based Dating Platform (Admin Portal) | *React, GraphQL, AWS Amplify*

- Built the admin portal for events, users, cities, and venues, with RBAC and multi-city scheduling.

MahaEseva.com – Online Rent Agreements | *Laravel, MySQL*

- Built admin and customer portals to create and e-sign government-compliant rent agreements across India, with online payments, biometric verification, RBAC, and lead management.

Software Developer | WebReinvent, New Delhi, India **Jul 2022 – Oct 2022**

- Built features and fixed bugs in VaahCMS, an open-source Laravel and Vue.js CMS (MySQL, REST APIs).

Full Stack Developer | **BluBrandz Technologies Pvt Ltd, New Delhi, India** **Aug 2021 – Jun 2022**

- Built Angle Fertility (hospital management: patient records, appointments, clinical workflows), Aragma (event management), and Unicorn Equipments (equipment tracking dashboard) with Laravel and MySQL.

PROJECTS AND OPEN SOURCE

Hopline – Encrypted Peer-to-Peer File Transfer | github.com/vishwajitvm/Hopline

Tech: *Node.js, WebRTC, WebSockets, React 19, Vite, Electron, AES-256-GCM, Docker, GitHub Actions*

- Built a peer-to-peer file transfer app for browser, desktop (Electron: Windows, macOS, Linux), and CLI with a Node.js WebSocket signaling server; live at hopline.onrender.com.
- Designed a WebRTC protocol that sends 64 KiB frames over up to 4 parallel connections, with flow control, SHA-256 verification, and resume after a disconnect.
- Built encrypted TCP transfer for CLI and desktop (CPace PAKE, HKDF, AES-256-GCM): **10 GiB in 56.8 s**, SHA-256 verified.
- Hardened the signaling server and TURN relay (rate limits, brute-force throttling); 90 automated tests, 4 security-test rounds fixed, and a 3-OS installer release pipeline.

DevFlow-AI – AI Code Review and CI/CD Platform | github.com/vishwajitvm/DevFlow-AI

Tech: *Python 3.13, FastAPI, SQLAlchemy 2, Celery, PostgreSQL, Redis, MCP, Ollama, Next.js 16, Docker, OpenTelemetry*

- Built a self-hosted AI code-review, auto-fix, and CI/CD platform for GitHub and GitLab: **132 API endpoints**, 31 tables, 30 Celery tasks, 17 Docker services.
- Built the Next.js 16 / React 19 / Tailwind console (live pipeline graphs, approvals, AI usage), covered by 43 Playwright tests.
- Wrote a framework-free tool-calling agent loop for 9 roles (analyst, fixer, code and security reviewers) with JSON-schema-validated output, token budgets, and AST caller-compatibility checks.
- Built an LLM router across 90 free models (10 providers) with 429 cool-downs and Ollama fallback, and an **MCP gateway** (57 tools) that gates every write with one-time keys and go / ask / stop rules.

Tallyko – Multi-Tenant Restaurant and Retail POS | github.com/vishwajitvm/Tallyko

- Built a multi-tenant POS (FastAPI, PostgreSQL row-level security, 38 tables, Redis) with an offline-first React Native (Expo) app that syncs queued bills without duplicates, local OCR menu import, and 400+ automated tests.

OmniShell – LLM-Planned Shell Automation with Human Approval | github.com/vishwajitvm/OmniShell

Tech: *Python, FastAPI, LiteLLM, Pydantic, PostgreSQL, Redis, NestJS, psutil, Docker*

- Built a FastAPI service that turns natural-language requests into OS workflows (Linux, macOS, Windows) across 19 capability types, using regex intent routing and one LiteLLM planning call with Pydantic-validated JSON.
- Designed a PostgreSQL scheduler with FOR UPDATE SKIP LOCKED claims, heartbeat leases, and SHA-256-hashed expiring approval tokens for human sign-off.

RepoMind – AI Codebase Q&A with RAG and LangGraph | github.com/vishwajitvm/RepoMind

- Built a LangGraph and FastAPI assistant that answers questions about GitHub repositories with file and line citations, using AST-based code chunking, Qdrant vector search, live GitHub API lookups, and 9-provider LLM fallback.

TenantMind-AI – Multi-Tenant AI Property Management | github.com/vishwajitvm/TenantMind-AI

- Built a multi-tenant FastAPI backend (database per tenant in MongoDB, Keycloak JWT), per-tenant Qdrant collections for document search, Celery workers, and an LLM gateway with fallback (Gemini, Groq, OpenRouter, Ollama) that logs latency and tokens per call.

AgentHive – Multi-Agent Orchestration Platform | github.com/vishwajitvm/AgentHive

- Built a ReAct agent platform (FastAPI, PostgreSQL, Celery, Next.js) with a database-configured LLM router (Gemini, OpenAI, Groq, Hugging Face, Ollama) with retry and fallback, 40 tools with per-agent allow-lists, and pgvector memory.

PolicyBot – RAG Q&A for Company Policies | github.com/vishwajitvm/PolicyBot

- Built a hybrid RAG pipeline (FastAPI, Qdrant, MongoDB, React) with vector and keyword retrieval, freshness-aware ranking, citations, and confidence scores, a LangGraph ingestion workflow, and per-stage traces for evaluation.

TraceNest – Python Logging SDK on PyPI | pypi.org/project/tracenest

- Published a zero-dependency, thread-safe logging SDK (18 releases) with daily rotation, recursive secret masking, FastAPI middleware, a built-in log viewer, and 42 tests; used in DevFlow-AI and OmniShell.

More AI Projects | github.com/vishwajitvm

- **LokTathya**: civic data platform for India (FastAPI, Next.js, PostgreSQL/PostGIS, 50 tables) with a GitHub Actions pipeline running migrations and 76 tests.
- **NL-Automation-Platform**: plain-English automations on 6 FastAPI microservices, with a LangGraph clarification agent and a fail-closed safety classifier.
- **FitTrack-AI**: meal nutrition from a photo via vision LLMs, grounded in USDA and Open Food Facts data (Next.js, FastAPI).

EDUCATION

Bachelor of Science (B.Sc.) | Panjab University, Chandigarh **2015 – 2018**